Earth and Planetary Science Letters, 87 (1988) 152–160
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

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Permo-Triassic magnetostratigraphy in China: northern Tarim

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Received January 20, 1987; revised version received October 9, 1987

The upper boundary of the Permo-Carboniferous Reversed Polarity Superchron has been identified in a palaeomagnetic study of the Permo-Triassic of the northern part of the Tarim Basin, China. This boundary serves as an important marker horizon for correlation with other Permo-Triassic sequences both in China and world-wide. A Permo-Triassic palaeomagnetic pole for the Tarim Block is estimated to be at 71.8° N, 187.6° E.

Comparison with similar age poles from the adjacent blocks of China and Asia suggests that the Tarim was widely separated from the Sino-Korean Block in Permo-Triassic times but was not yet sutured to Kazakhstan.

1. Introduction

It is now widely recognised that Asia is a composite continent formed by the accretion of various crustal blocks during the Palaeozoic and Mesozoic [1–3]. Analyses of palaeomagnetic data have confirmed this to be the case [4–6] and currently much work is being done to understand the tectonic history of the various palaeoblocks, especially in China.

Fig. 1 shows a subdivision of China into palaeoblocks that once existed as separate plates

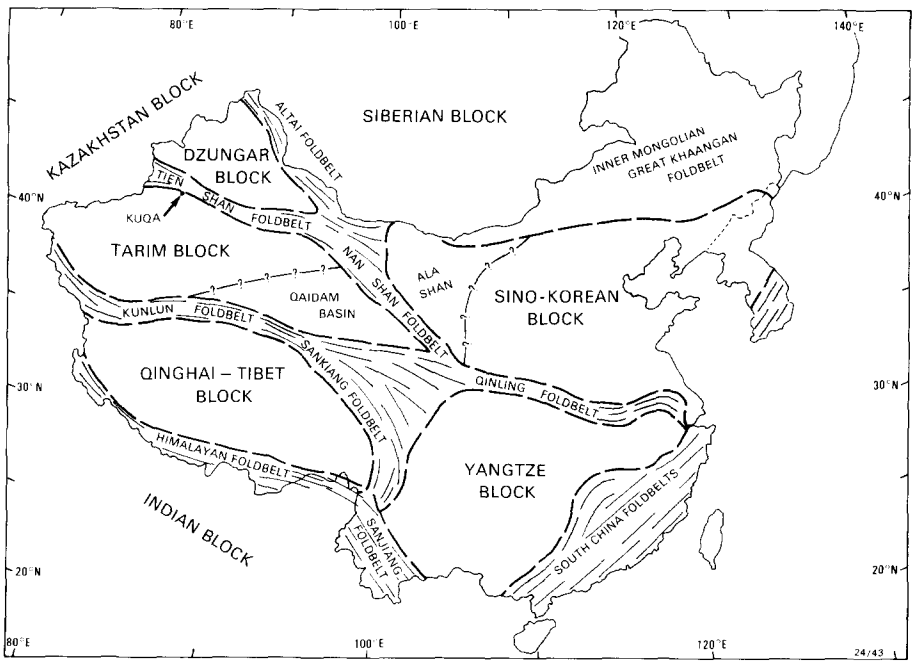


Fig. 1. Generalized tectonic map of China. The palaeomagnetic sampling locality in the northern Tarim Block near Kuqa is indicated.

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[3,7,8]. There is currently much activity in palaeomagnetic studies of the Sino-Korean and Yangtze Blocks of China, but very little work has been carried out in the Tarim Block except for some recent Palaeozoic studies carried out by Chinese workers [9] from the State Seismological Bureau. We report here some results from the Permo-Triassic of the sequence exposed in the northern part of the Tarim Block in the southern part of the Tien Shan Mountains near Kuqa. This program is part of a continuing series of co-operative palaeomagnetic investigations of the Permo-Triassic of China between the Chinese Academy of Geological Sciences and Australian palaeomagnetists.

A major purpose of the study was to identify the upper boundary of the Permo-Carboniferous Reversed Polarity Superchron [10] in this part of China. In this way correlation can be made with the similar sequence sampled in the type section of the Sino-Korean Block near Taiyuan, Shansi Province [11].

2. Sampling

The type section of the Permo-Triassic of the northern part of the Tarim Basin is situated north of Kuqa in the Tien Shan Mountains at Biyulopaokutze (42°08′N, 83°21′E). The sequence here is estimated to be 870 m thick in layers numbered 1 to 54 starting at the base. Each of these layers consists of several bands ranging up to 50 m thick typically consisting of conglomerates with interbedded mudstones and sandstones. Layers 1–22 covering the first 350 m of the section are largely greenish sandstones and mudstones with conglomerates and are probably Lower to mid-Permian. Layer 23 is a very thick ($\simeq$ 30 m) conglomerate that marks the start of an important change in the lithology, from mainly greenish in character, to red and purple sediments that are more suitable for palaeomagnetic studies.

The present study was carried out as part of a wide-spread investigation of Permo-Triassic sequences in China using palaeontology and magnetostratigraphy as a means of correlating these sequences precisely across the country. The palaeontological studies, still to be published, identify the Permo-Triassic boundary as lying at about Layer 40 in this sequence. The identification of the upper boundary of the Permo-Carboniferous Reversed Polarity Superchron [10] in the sequence would thus play an important rôle in precise time-stratigraphic correlation. Fortuitously this boundary, on the palaeontological evidence, occurs in a sequence with some of the most favourable attributes for palaeomagnetic investigations.

From Layer 26 upwards the sequence consists of many red, purple and chocolate sandstones, shales and mudstones with intervening conglomerates. The upper Permian sequence from Layers 26–39 is about 200 m thick with the overlying Lower Triassic sequence from Layers 40–54 being 280 m thick. The entire section dips to the west at about 10°. In total, 25 horizons were sampled in this section, providing a collection of 172 samples. At the base of the section only Layers 2, 4, 8 and 10 were sampled because of the nature of the material. More detailed collections were made from Layers 26–47 where the sediments appeared ideal for palaeomagnetic studies.

A more complex section from a structural viewpoint was also sampled at Ohuobulhake (42°14′N, 82°59′E). The section, about 240 m thick, consists of Upper Permian rocks overturned by the upthrusting of some older rocks and a granite. A thick conglomerate near the base is regarded as the equivalent of Layer 23 in the Biyulopaokutze section and this is the main reason why the sequence is regarded as Upper Permian. Twelve horizons were sampled involving a collection of 64 samples.

3. Palaeomagnetic measurements and analysis

The samples were processed in the palaeomagnetic laboratories at Canberra and Beijing, roughly 50% at each. All specimens were subjected to step-wise thermal demagnetization up to 680°C. The thermal demagnetization equipment set up in the Beijing laboratory is virtually identical to that in use in Canberra [12]. In Canberra all measurements were made on a two-axis cryogenic SQUID magnetometer and in Beijing a DIGICO magnetometer was used.

Specimens from layers with normal polarity either contained only one component of magnetization (apart from any temporary components removed in the first 200°C, see Fig. 2a) or were

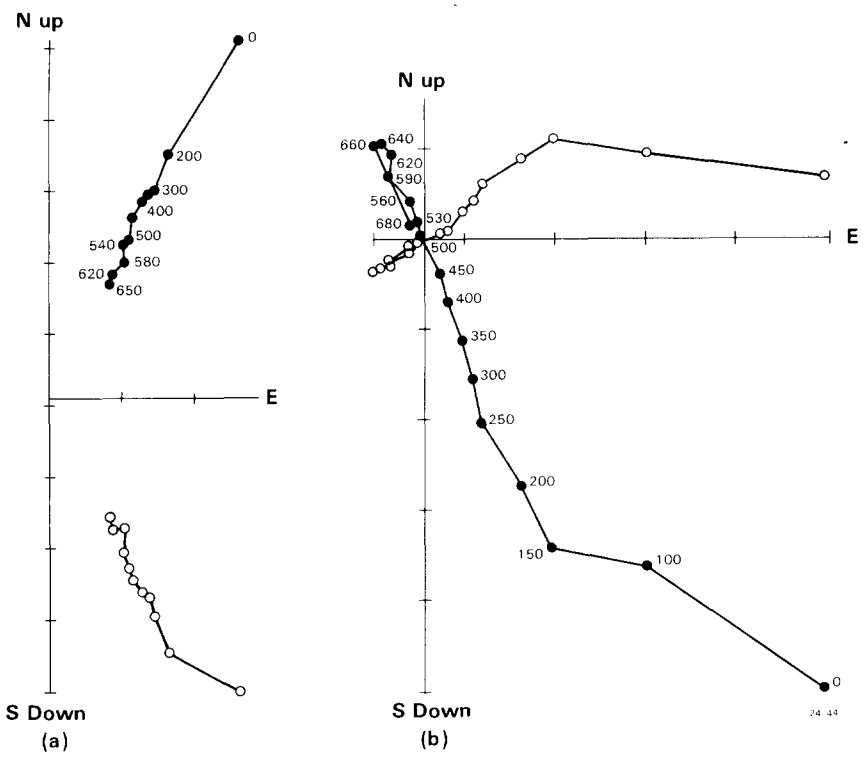


Fig. 2. Zijderveld diagrams illustrating the thermal demagnetization characteristics of typical specimens dominated by a single stable component of magnetization. (a) Normally magnetized specimens, (b) reversely magnetized specimens. Solid circles in horizontal plane, open circles in vertical plane. Temperatures are indicated for each point in °C.

overprinted by the present-day field. These overprints were generally removed, as judged by an analysis of the removed vectors in each case, during thermal demagnetization. However, specimens from layers with reversed polarity were heavily overprinted, the primary magnetization being retained only by those grains with blocking temperatures close to the Curie point of haematite. In only a few cases was a direct observation (stable endpoint) made of the underlying characteristic component of magnetization as judged from the analysis of a linear segment near the origin of a Zijderveld plot (see Fig. 2b). The present field direction lies roughly 160° away from the reversed characteristic magnetization. Thus successive demagnetizations typically moved the vector observed along a great circle towards that characteristic direction. In many instances however the characteristic magnetization was never reached because of the necessity to make measurements successively within a few degrees of the Curie Point. Unfortunately the residual magnetization usually became very weak near the Curie Point of hematite. In our analyses we have noted that at low intensities the directions of magnetization when measured on the Digico magnetometer in Beijing were more scattered than those measured on the Squid magnetometer in Canberra, as was to be expected. This has led to complications in trying to analyse and combine the sets of data from the two laboratories.

Thus there is no linear segment near the origin of the Zijderveld plot that is capable of analysis by standard packages such as LINEFIND [13] or Linearity Spectrum Analysis (LSA) [14]. An example is given in Fig. 3. The standard Zijderveld plot in Fig. 3b shows that the final vector magnetization has not been observed. If the directions of magnetization at successive temperatures are plotted on a stereographic projection as shown in Fig. 3a, it is apparent that the "endpoint" direction should be constrained to lie along the sector of the great circle somewhere just before the last high-temperature measurement (because of the

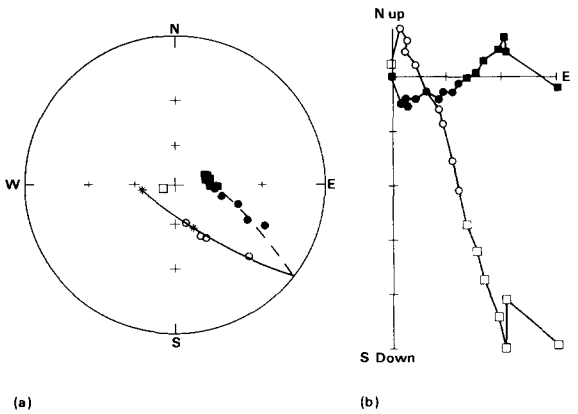


Fig. 3. An example of the demagnetization characteristics of a specimen for which the vector moves along a great circle but never reaches the underlying stable direction (as judged by a linear segment going towards the origin of the Zijderveld plot). Circles indicate those points used to define the great circle, squares indicate the other points at low or high temperatures that are not a part of the great circle. (a) Stereographic projection showing directions of magnetization at successive temperatures with solid symbols (dashed great circle) on the lower hemisphere and open symbols (unbroken great circle) on the upper hemisphere. The best estimate of the underlying direction of magnetization is constrained to be along the great circle between the two starred points (see companion paper [15] for details). (b) Zijderveld plots for vectors displayed in (a). Solid circles lie in the horizontal plane, open circles lie in the vertical plane.

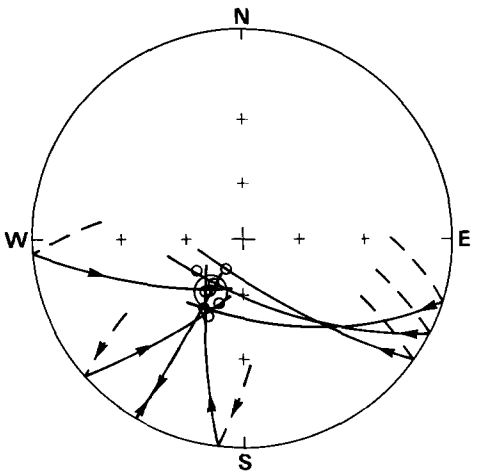


Fig. 4. An example of the analysis of 9 specimens from Layer 28. The data comprise two fixed points (squares) and seven great circles. Each of the great circles terminates at the point defined by the limiting sector constraint (usually 180° from the initial direction). The open circles are the best estimates of the directions of magnetization on the great circle arcs determined by the interactive process described in the text and in the companion paper [15]. Stereographic projection with open symbols (unbroken great circle) on the upper hemisphere. The circle of 95% confidence for the mean is shown.

TABLE 1

Directions of magnetization observed in the Permo-Triassic sequence near Biyulopaokutze, Tarim Block, China (location 42°08′N, 83°21′E)

Layer	N (n, c)	D (°)	I (°)	R	k	α_{95} (°)
47	3 (2, 1)	212.7	−56.8	2.970	50.0	20.7
44	7 (0, 7)	226.2	−48.7	6.776	11.2	22.6
43	11 (0, 11)	216.4	−56.8	10.805	23.1	10.6
42	8 (0, 8)	200.0	−43.5	7.898	29.4	12.1
41B	6 (0, 6)	169.6	−64.0	5.632	5.4	39.4
41A	6 (6, 0)	17.2	54.6	5.920	62.4	8.6
37	4 (4, 0)	12.7	63.0	3.681	9.4	31.6
36	8 (8, 0)	26.2	55.8	7.395	11.6	17.0
35	5 (5, 0)	22.3	48.7	4.967	120.4	7.0
34	4 (1, 3)	175.3	−68.3	3.954	32.6	22.2
32	7 (0, 7)	212.1	−45.1	6.827	14.5	19.7
31	8 (2, 6)	184.8	−51.7	7.833	23.9	12.5
29	4 (0, 4)	193.6	−65.0	3.999	1000	5.6
28	9 (2, 7)	212.0	−57.1	8.899	44.6	8.4
27	6 (0, 6)	214.7	−40.2	5.886	17.5	21.1
26	7 (2, 5)	187.6	−45.4	6.927	47.9	9.7
Early Triassic (41–47)	6	205.4	−55.2	5.8749	40.0	10.7
Late Permian (26–37)	10	199.7	−54.7	9.8007	45.2	7.3
Combined Permo-Triassic	16	201.8	−54.9	15.6694	45.4	5.5

Pole-Positions	N	Lat. °N	Long. °E	dp, dm
Early Triassic	6	69.3	182.9	10.8, 15.2
Late Permian	10	73.2	191.0	7.3, 10.3
Permo-Triassic	16	71.8	187.6	5.5, 7.8

D = declination, I = inclination, N = number of samples, n = number of direct observations, c = number of remagnetization circles, R = resultant of N unit vectors, k = Fisher's precision parameter, α_{95} = radius of circle of 95% confidence about the mean.

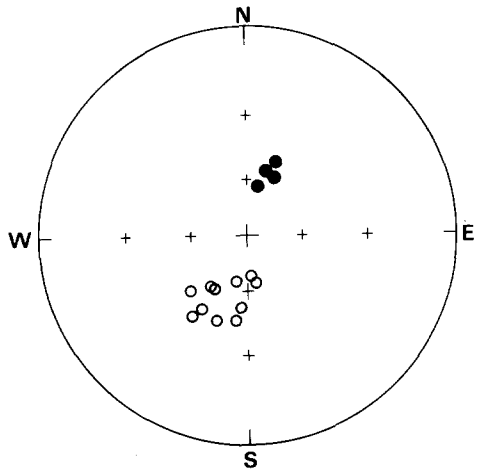


Fig. 5. Stereographic projection of the mean directions of magnetization determined from 16 layers giving useful palaeomagnetic results. Solid circles on the lower hemisphere, open circles on the upper hemisphere.

statistical precision with which that last measurement has been determined) and, in this instance, a point no further than 180° away from the initial room-temperature measurement. The great circle in Fig. 3a has been terminated at this point.

A technique for the combined analysis of remagnetization circles and direct observations (fixed directions) using the sector constraints described above, has been developed in a companion paper [15]. An iterative procedure is used to determine the maximum likelihood estimate of the endpoint direction within the acceptable sector for each great circle. When these are combined with the fixed directions using Fisher [16] statistics, the best estimate of the mean direction is determined. The method has the advantage of eliminating bias that is inherent in other analyses of converging great circles [17].

An example of such an analysis is given in Fig. 4. The companion paper [15] provides a further example. Statistical information for the mean direction determined for each layer is provided in Table 1 and these are illustrated in Fig. 5 for the 16 layers providing palaeomagnetic information.

A change of polarity from normal to reversed occurs within Layer 41, hence two separate directions have been recorded for that layer. Results could only be obtained from the sequence at Biyulopaokutze. The structural complexity of the sequence at Ohuobulhake appears to have caused massive overprinting of the rocks in that region and no useful palaeomagnetic information was obtained.

It is now becoming widely accepted that red beds may carry ancient and stable, yet secondary, magnetizations without any trace of the primary magnetization. The normally magnetized characteristic directions observed in the sequence from Layers 35 to 41 lie close to the axial dipole field direction (within 20°). This means that the large normal overprints are close to being parallel or

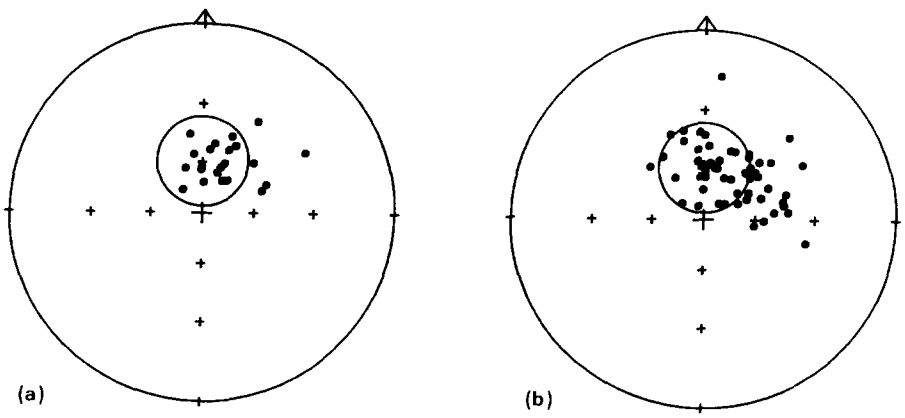


Fig. 6. Removed vectors (up to 400° C) from individual samples (a) with normally magnetized characteristic directions, and (b) with reversely magnetized directions. In each case the circle is centred around the axial dipole field direction (declination 0.0°, inclination 61.1°) with a semi-angle of 25°, and shows the region within which about 95% of directions should lie if the observed scatter were due to expected Brunhes-age palaeosecular variation.

antiparallel to the characteristic magnetization. The most important question to be determined is therefore the age of the overprint. There appear to be two possibilities for this.

(1) All the beds once had the reversed direction but the normal (probably very early) remagnetization affected the whole section, completely remagnetizing four beds in the middle of the section and strongly overprinting the others.

(2) The overprint is a Brunhes-age axial dipole component on top of a primary Permo-Triassic magnetization.

In an attempt to distinguish between these two possibilities we have analysed the removed vectors calculated at each step of demagnetization. It has already been mentioned that there were problems with the accuracy of measurement at low intensities on the Digico magnetometer, so the analysis presented covers specimens measured on the Squid magnetometer in Canberra. These removed vectors (up to 400 °C) are illustrated in Fig. 6 separately for the samples with normally magnetized characteristic directions and those with reversely magnetized directions. Our interpretation of these directions is that the removed vectors are a combination of a set of directions scattered about the axial dipole field direction and the characteristic direction in each case. To illustrate this a circle of semi-angle 25° has been drawn around the axial dipole field direction to show the region on the stereonet within which about 95% of directions should lie if the observed scatter were due to expected Brunhes-age palaeosecular variation [20–22] at latitude 42°. In Fig. 6a for the normal directions, most removed vectors are contained symmetrically within the 25° circle with a few outside to the right, as would be expected as a result of contamination from the removal of some of the characteristic normal magnetization (with $D = 20°$). In Fig. 6b for the reversed directions, again with the 25° circle the removed vectors are symmetrically distributed but are strung out towards the reversed characteristic direction, again as would be expected.

Naturally the above arguments do not prove that this is the correct interpretation of the removed vectors, but the observations are consistent with such an interpretation. Perhaps the most compelling argument is the consistency of the observed magnetostratigraphy from the characteristic directions with that to be expected from the stratigraphic position of the sequence deduced from the palaeontology (see Fig. 7). On this basis it is argued that the characteristic directions of magnetization observed are most likely original primary magnetizations acquired when the sediments were deposited.

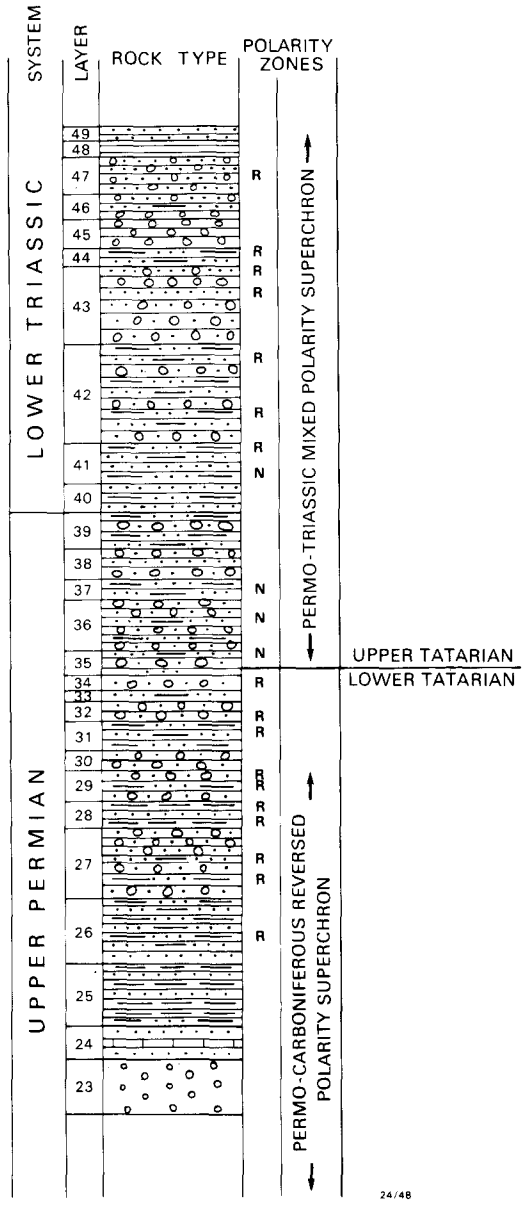


Fig. 7. Magnetostratigraphy determined for the Permo-Triassic sequence at Biyulopaokutze near Kuqa, northern Tarim Block.

4. Magnetostratigraphy

Fig. 7 provides a description of the polarity sequence obtained from the results at Biyulopaokutze. From Layers 26 through 34 the sequence exhibits consistent reversed polarity followed by a normal zone from Layer 35 to the base of Layer 41. Above this the sequence is reversely magnetized up to Layer 47. Accepting the location of the Permo-Triassic boundary at the base of Layer 40, the palaeomagnetic information is consistent with the upper boundary of the Permo-Carboniferous Reversed Polarity Superchron [10,18] occurring at the base of Layer 35.

A very similar result has been obtained from the standard Permo-Triassic sequence in the Sino-Korean block near Taiyuan [11]. Together these form the basis of a Permo-Triassic magnetostratigraphy not only for China but world-wide. The results are consistent with current information from the Soviet Union, western Europe and the United States [10,18], identifying the upper boundary of the Permo-Carboniferous Reversed Polarity Superchron as occurring in the mid to late Tatarian.

5. Palaeogeography

The palaeomagnetic pole determined for the Permo-Triassic sequence in northern Tarim is

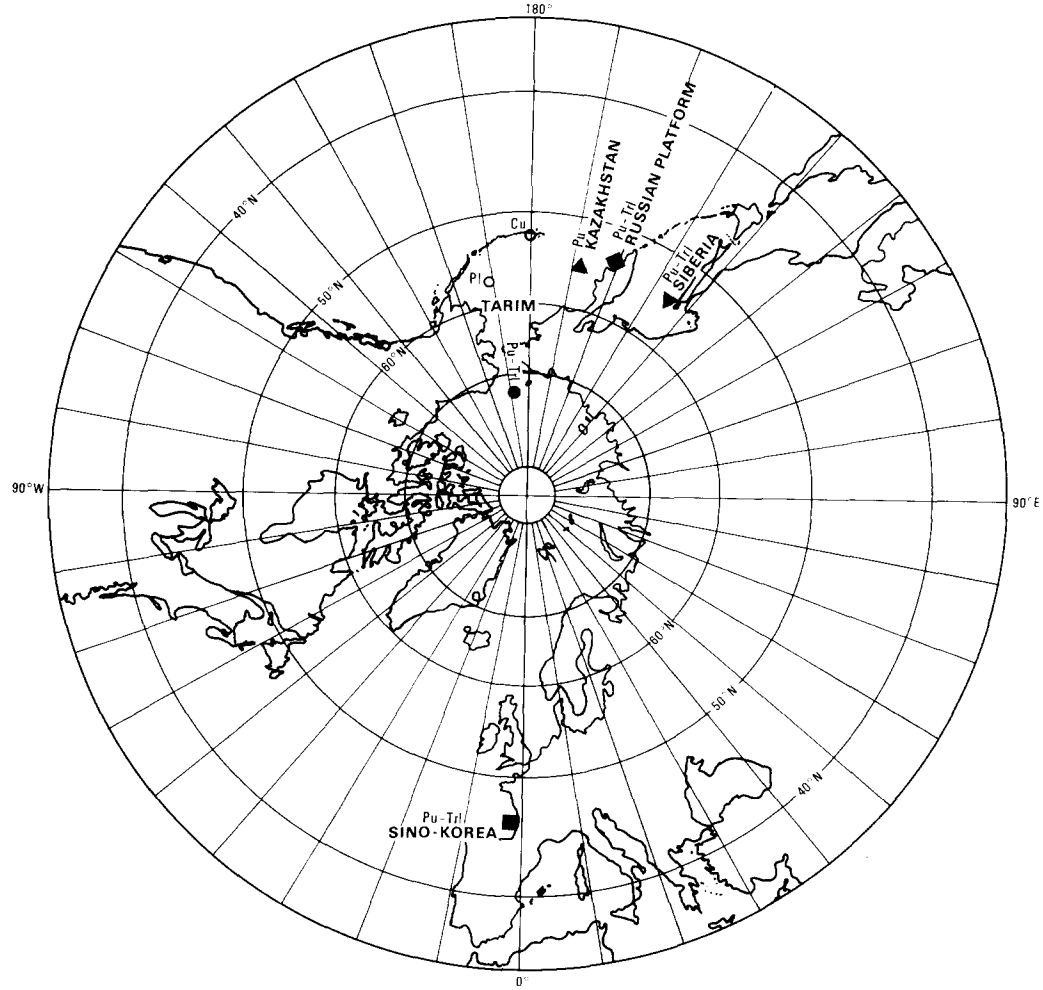


Fig. 8. Permo-Triassic palaeomagnetic poles for Siberia, Kazakhstan and the Russian Platform [19] compared with those from the Sino-Korean Platform [11] and the Tarim Block (this paper). Permo-Carboniferous poles from Tarim [9] are also shown for comparison.

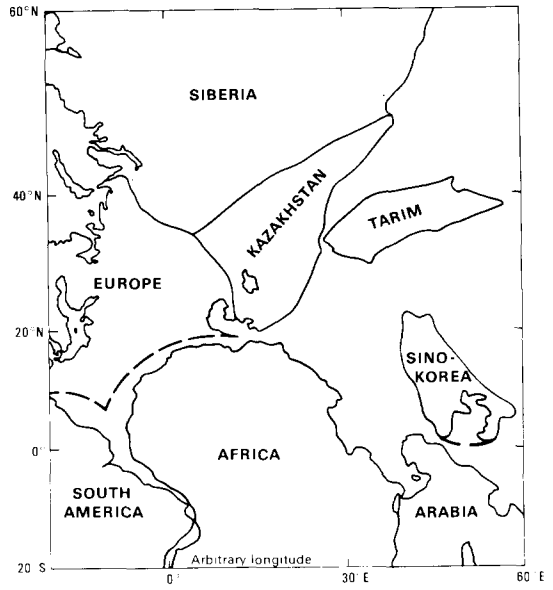


Fig. 9. Permo-Triassic palaeogeographic map showing possible positions of the Tarim and Sino-Korean Blocks with respect to a reconstruction of Gondwana and Laurasia. Relative longitudes are indeterminate.

situated at 72° N, 188° E (Table 1). It is significantly displaced (20° of arc) from the group of Permo-Triassic poles for Kazakhstan, Siberia and the Russian platform as determined by Khramov et al. [19] (see Fig. 8). It is very widely displaced from the Late Permian pole determined for the Sino-Korean Block [11]. These comparisons suggest that by Permo-Triassic times the Tarim Block was not yet sutured to the Kazakhstan Block contrary to the conclusion drawn by Bai et al. [9] from late Palaeozoic data for Tarim observed in the Aksu area. This conclusion was based on the observation that Late Carboniferous and Early Permian poles for Tarim were reasonably close to those for the Permo-Triassic of Kazakhstan, Siberia and the Russian platform (see Fig. 8). Our information for the Permo-Triassic shows that this conclusion is not tenable.

Fig. 9 shows the possible palaeogeography and relationships between Tarim, Sino-Korea, Laurasia and Gondwana in Permo-Triassic times. Tarim was probably very close to Laurasia but not yet sutured to the Kazakhstan Block. On the other hand Sino-Korea may have been associated with the northern part of Gondwana as proposed by Lin et al. [6].

Acknowledgements

We wish to thank the Geological Bureau of the Xinjiang Autonomous Region for their help with this project and especially Wu Shao Zu for his assistance in the field. This paper is published with the Permission of the Director of the Bureau of Mineral Resources, Geology and Geophysics.

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